Partial Fractions

ANSWER KEY

Distinct linear factors

1 Express $\dfrac{5x+7}{(x+1)(x+2)}$ in partial fractions.
Write $\dfrac{5x+7}{(x+1)(x+2)} = \dfrac{A}{x+1} + \dfrac{B}{x+2}$. Multiply through: $5x+7 = A(x+2) + B(x+1)$. At $x = -1$: $2 = A(1)$, so $A = 2$. At $x = -2$: $-3 = B(-1)$, so $B = 3$. Result: $\dfrac{2}{x+1} + \dfrac{3}{x+2}$.

2 Express $\dfrac{3x-1}{x(x-3)}$ in partial fractions.
Write $\dfrac{3x-1}{x(x-3)} = \dfrac{A}{x} + \dfrac{B}{x-3}$. Then $3x-1 = A(x-3) + Bx$. At $x = 0$: $-1 = -3A$, so $A = \dfrac{1}{3}$. At $x = 3$: $8 = 3B$, so $B = \dfrac{8}{3}$. Result: $\dfrac{1}{3x} + \dfrac{8}{3(x-3)}$.

Repeated linear factors

3 Express $\dfrac{4x+1}{(x-1)^2}$ in partial fractions of the form $\dfrac{A}{x-1} + \dfrac{B}{(x-1)^2}$.
$4x+1 = A(x-1) + B$. At $x = 1$: $5 = B$. Comparing coefficients of x: $A = 4$. Result: $\dfrac{4}{x-1} + \dfrac{5}{(x-1)^2}$.

4 Express $\dfrac{2x^2-3x+1}{x(x-1)^2}$ in partial fractions of the form $\dfrac{A}{x} + \dfrac{B}{x-1} + \dfrac{C}{(x-1)^2}$.
$2x^2-3x+1 = A(x-1)^2 + Bx(x-1) + Cx$. At $x = 0$: $1 = A(1)$, so $A = 1$. At $x = 1$: $0 = C(1)$, so $C = 0$. Comparing x^2 coefficients: $2 = A + B$, so $B = 2 - 1 = 1$. Result: $\dfrac{1}{x} + \dfrac{1}{x-1}$.

Irreducible quadratic factors

5 Express $\dfrac{3x^2+2}{(x-1)(x^2+1)}$ in partial fractions of the form $\dfrac{A}{x-1} + \dfrac{Bx+C}{x^2+1}$.
$3x^2+2 = A(x^2+1) + (Bx+C)(x-1)$. At $x = 1$: $5 = 2A$, so $A = \dfrac{5}{2}$. Comparing x^2 coefficients: $3 = A + B$, so $B = 3 - \dfrac{5}{2} = \dfrac{1}{2}$. Comparing constants: $2 = A - C$, so $C = A - 2 = \dfrac{1}{2}$. Result: $\dfrac{5}{2(x-1)} + \dfrac{x+1}{2(x^2+1)}$.

Applying partial fractions to integration

6 Use partial fractions to evaluate $\displaystyle\int \frac{5x+7}{(x+1)(x+2)}\,dx$, using the decomposition found earlier in this worksheet.
From earlier: $\dfrac{5x+7}{(x+1)(x+2)} = \dfrac{2}{x+1} + \dfrac{3}{x+2}$. So $\displaystyle\int \frac{5x+7}{(x+1)(x+2)}\,dx = 2\ln|x+1| + 3\ln|x+2| + C$.

Applying partial fractions to a binomial series expansion

7 Express $\dfrac{1}{(1-x)(1-2x)}$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x.

$\dfrac{1}{(1-x)(1-2x)} = \dfrac{A}{1-x} + \dfrac{B}{1-2x}$. Then $1 = A(1-2x) + B(1-x)$. At $x = 1$: $1 = A(-1)$, so $A = -1$. At $x = \frac{1}{2}$: $1 = B\left(\frac{1}{2}\right)$, so $B = 2$. Result: $\dfrac{-1}{1-x} + \dfrac{2}{1-2x}$. Expanding each geometric series: $-1(1 + x + x^2 + \cdots) + 2(1 + 2x + 4x^2 + \cdots) = (-1+2) + (-1+4)x + (-1+8)x^2 + \cdots = 1 + 3x + 7x^2 + \cdots$.

Combining terms before decomposing

8 Express $\dfrac{7-x}{(x+1)(x-2)}$ in partial fractions.

Write $\dfrac{7-x}{(x+1)(x-2)} = \dfrac{A}{x+1} + \dfrac{B}{x-2}$. Then $7 - x = A(x-2) + B(x+1)$. At $x = -1$: $8 = A(-3)$, so $A = -\frac{8}{3}$. At $x = 2$: $5 = B(3)$, so $B = \frac{5}{3}$. Result: $\dfrac{-8}{3(x+1)} + \dfrac{5}{3(x-2)}$.

9 Express $\dfrac{x^2+x+1}{x(x+1)}$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)

Since numerator and denominator are both degree 2, divide first: $\dfrac{x^2+x+1}{x^2+x} = 1 + \dfrac{1}{x^2+x} = 1 + \dfrac{1}{x(x+1)}$. Decompose $\dfrac{1}{x(x+1)} = \dfrac{A}{x} + \dfrac{B}{x+1}$: $1 = A(x+1) + Bx$. At $x = 0$: $A = 1$. At $x = -1$: $B = -1$. Result: $1 + \frac{1}{x} - \dfrac{1}{x+1}$.

Applying partial fractions to a rate problem

10 A population model gives the rate of change $\dfrac{dP}{dt} = \dfrac{1}{P(10-P)}$. Express $\dfrac{1}{P(10-P)}$ in partial fractions of the form $\dfrac{A}{P} + \dfrac{B}{10-P}$.

$1 = A(10-P) + BP$. At $P = 0$: $1 = 10A$, so $A = \dfrac{1}{10}$. At $P = 10$: $1 = 10B$, so $B = \dfrac{1}{10}$. Result: $\dfrac{1}{10P} + \dfrac{1}{10(10-P)}$ — a decomposition commonly used to solve logistic-growth differential equations by separation of variables.

Partial fractions with three distinct linear factors

11 Express $\dfrac{6x^2-2x-1}{(x-1)(x+1)(x-2)}$ in partial fractions.

Write $\dfrac{6x^2-2x-1}{(x-1)(x+1)(x-2)} = \dfrac{A}{x-1} + \dfrac{B}{x+1} + \dfrac{C}{x-2}$. Then $6x^2 - 2x - 1 = A(x+1)(x-2) + B(x-1)(x-2) + C(x-1)(x+1)$. At $x = 1$: $3 = A(2)(-1) = -2A$, so $A = -\dfrac{3}{2}$. At $x = -1$: $7 = B(-2)(-3) = 6B$, so $B = \dfrac{7}{6}$. At $x = 2$: $19 = C(1)(3) = 3C$, so $C = \dfrac{19}{3}$.

Partial fractions with an improper fraction and repeated factor

12 Express $\frac{x^3}{(x-1)^2}$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.

Divide x^3 by $(x-1)^2 = x^2 - 2x + 1$: $x^3 \div (x^2 - 2x + 1) = x + 2$ remainder $3x - 2$. So $\frac{x^3}{(x-1)^2} = x + 2 + \frac{3x-2}{(x-1)^2}$. Decompose the remainder: $\frac{3x-2}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, giving $3x - 2 = A(x-1) + B$. At $x = 1$: $1 = B$. Comparing x coefficients: $A = 3$. Result: $x + 2 + \frac{3}{x-1} + \frac{1}{(x-1)^2}$.

Using partial fractions to sum a series

13 Express $\frac{1}{n(n+1)}$ in partial fractions, and use the result to find $\sum_{n=1}^{4} \frac{1}{n(n+1)}$ by telescoping.
$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ (standard decomposition with $A = 1$, $B = -1$). Summing from $n = 1$ to 4: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) = 1 - \frac{1}{5} = \frac{4}{5}$, since all the middle terms cancel (telescope).

Decomposing a fraction with a repeated quadratic-adjacent structure

14 Express $\frac{x+5}{(x-1)(x+1)^2}$ in partial fractions of the form $\frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$.
$x + 5 = A(x+1)^2 + B(x-1)(x+1) + C(x-1)$. At $x = 1$: $6 = A(4)$, so $A = \frac{3}{2}$. At $x = -1$: $4 = C(-2)$, so $C = -2$. Comparing x^2 coefficients: $0 = A + B$, so $B = -\frac{3}{2}$. Result: $\frac{3}{2(x-1)} - \frac{3}{2(x+1)} - \frac{2}{(x+1)^2}$.

15 Express $\frac{2x^2 + x - 1}{x^2(x+1)}$ in partial fractions of the form $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$.
$2x^2 + x - 1 = Ax(x+1) + B(x+1) + Cx^2$. At $x = 0$: $-1 = B(1)$, so $B = -1$. At $x = -1$: $0 = C(1)$, so $C = 0$. Comparing x^2 coefficients: $2 = A + C = A$, so $A = 2$. Result: $\frac{2}{x} - \frac{1}{x^2}$.

16 Use partial fractions to find the Maclaurin (power series) expansion of $\frac{1}{(1-x)(2-x)}$ up to the term in x^2, first decomposing into the form $\frac{A}{1-x} + \frac{B}{2-x}$.
$1 = A(2-x) + B(1-x)$. At $x = 1$: $1 = A(1)$, so $A = 1$. At $x = 2$: $1 = B(-1)$, so $B = -1$. Result: $\frac{1}{1-x} - \frac{1}{2-x} = \frac{1}{1-x} - \frac{1}{2} \cdot \frac{1}{1-x/2}$. Expanding each as a geometric series: $(1 + x + x^2 + \cdots) - \frac{1}{2}\left(1 + \frac{x}{2} + \frac{x^2}{4} + \cdots\right) = \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{4}\right)x + \left(1 - \frac{1}{8}\right)x^2 + \cdots = \frac{1}{2} + \frac{3}{4}x + \frac{7}{8}x^2 + \cdots$.

17 Express $\frac{4x^2 - 5x - 15}{(x+1)(x-2)(x+3)}$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).
Write $\frac{4x^2 - 5x - 15}{(x+1)(x-2)(x+3)} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{x+3}$. Using the cover-up method:
$A = \frac{4(-1)^2 - 5(-1) - 15}{(-1-2)(-1+3)} = \frac{4 + 5 - 15}{(-3)(2)} = \frac{-6}{-6} = 1$.
$B = \frac{4(2)^2 - 5(2) - 15}{(2+1)(2+3)} = \frac{16 - 10 - 15}{(3)(5)} = \frac{-9}{15} = -\frac{3}{5}$.
$C = \frac{4(-3)^2 - 5(-3) - 15}{(-3+1)(-3-2)} = \frac{36 + 15 - 15}{(-2)(-5)} = \frac{36}{10} = \frac{18}{5}$. Result: $\frac{1}{x+1} - \frac{3}{5(x-2)} + \frac{18}{5(x+3)}$.

18 A chemical reaction's rate is modeled by $\frac{dC}{dt} = \frac{1}{(C+2)(C-1)}$, where C is a concentration. Express $\frac{1}{(C+2)(C-1)}$ in partial fractions of the form $\frac{A}{C+2} + \frac{B}{C-1}$, and then use the result to write the general antiderivative $\int \frac{1}{(C+2)(C-1)}\, dC$ in terms of natural logarithms.
$1 = A(C-1) + B(C+2)$. At $C = 1$: $1 = B(3)$, so $B = \frac{1}{3}$. At $C = -2$: $1 = A(-3)$, so $A = -\frac{1}{3}$. Result: $\frac{-1/3}{C+2} + \frac{1/3}{C-1}$. Integrating: $\int \frac{1}{(C+2)(C-1)}\, dC = -\frac{1}{3}\ln|C+2| + \frac{1}{3}\ln|C-1| + K = \frac{1}{3}\ln\left|\frac{C-1}{C+2}\right| + K$.

19 Use the decomposition $\frac{1}{3x} + \frac{8}{3(x-3)}$ found earlier in this worksheet for $\frac{3x-1}{x(x-3)}$ to evaluate $\int_4^5 \frac{3x-1}{x(x-3)}\,dx$, giving your answer as a single logarithm expression.

$\int \frac{3x-1}{x(x-3)}\,dx = \frac{1}{3}\ln|x| + \frac{8}{3}\ln|x-3| + C$. Evaluating from 4 to 5: $\left[\frac{1}{3}\ln 5 + \frac{8}{3}\ln 2\right] - \left[\frac{1}{3}\ln 4 + \frac{8}{3}\ln 1\right] = \frac{1}{3}\ln 5 + \frac{8}{3}\ln 2 - \frac{1}{3}\ln 4 = \frac{1}{3}\ln\frac{5}{4} + \frac{8}{3}\ln 2$.

20 This question has three parts. (a) Factor $x^2 - 4$ and hence express $\frac{x+8}{x^2-4}$ in partial fractions. (b) Use your result from part (a) to write down $\int \frac{x+8}{x^2-4}\,dx$. (c) Hence evaluate $\int_3^5 \frac{x+8}{x^2-4}\,dx$, giving your answer as a single expression involving natural logarithms.

(a) Factor $x^2 - 4 = (x-2)(x+2)$, so $\frac{x+8}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$. Then $x+8 = A(x+2) + B(x-2)$. At $x = 2$: $10 = 4A$, so $A = \frac{5}{2}$. At $x = -2$: $6 = -4B$, so $B = -\frac{3}{2}$. Result: $\frac{5/2}{x-2} - \frac{3/2}{x+2}$. (b) $\int \frac{x+8}{x^2-4}\,dx = \frac{5}{2}\ln|x-2| - \frac{3}{2}\ln|x+2| + C$. (c) Evaluating from 3 to 5: $\left[\frac{5}{2}\ln 3 - \frac{3}{2}\ln 7\right] - \left[\frac{5}{2}\ln 1 - \frac{3}{2}\ln 5\right] = \frac{5}{2}\ln 3 - \frac{3}{2}\ln 7 + \frac{3}{2}\ln 5$ (since $\ln 1 = 0$).